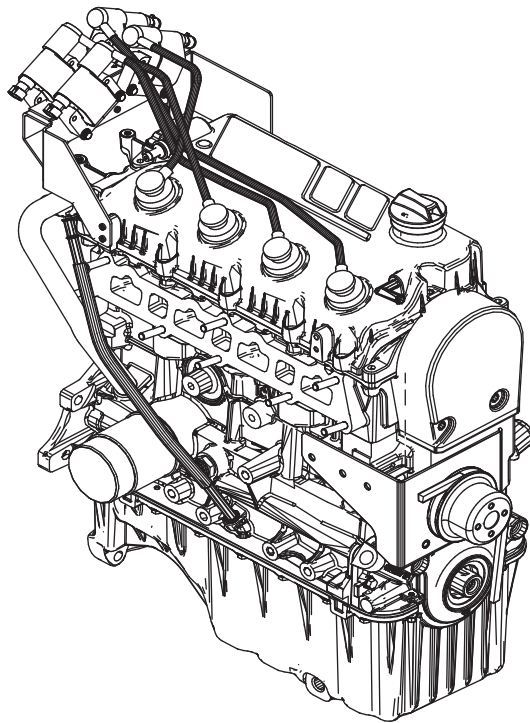




1.5L Gaseous Engine (G11)

Service Manual

*For
Generators*



NOTICE!

The information in this manual is known to be accurate as of the date of release shown on the cover. However, engineering changes can occur without notice. To obtain the proper parts, please be sure to use the serial number of the unit being serviced when ordering from this manual. Always refer to any part notes and changes before ordering. All other parts not shown in this manual can be found on Generac's exploded view drawings online on GENservice at www.generac.com.

SAVE THIS MANUAL FOR FUTURE REFERENCE

⚠ WARNING

California Proposition 65. Engine exhaust and some of its constituents are known to the state of California to cause cancer, birth defects, and other reproductive harm.

(000004)

⚠ WARNING

California Proposition 65. This product contains or emits chemicals known to the state of California to cause cancer, birth defects, and other reproductive harm.

(000005)

Table of Contents

Section 1 Introduction and Safety

Introduction	1
Safety Rules	1
Safety Symbols and Meanings	2

Section 2 General Information

Engine Information	3
Engine Identification	3
Parts Information	3
Specifications	4

Section 3 Assembly and Installation

Assemble Cylinder Block	5
Grouping and Mark Positions of Cylinder Bores	5
Install Main Bearing Cap	5
Install Main Bearing Cap Bolts	5
Press-Fit of Bowl Plug	5
Install Inner Hexagon Tapered Plug	6
Assemble Piston	6
Assemble Connecting Rod	6
Install Piston Ring	7
Assemble Oil Baffle	7
Install Main Bearing Shell	7
Install Connecting Rod Bearing Shell	7
Install Crankshaft, Crankshaft Woodruff Key and Crankshaft Thrust Plate	7
Install Oil Pump, Crankshaft Front Oil Seal and Oil Strainer	8
Assemble Front Oil Seal	8
Install Oil Pump with Front Oil Seal Assembly	8
Install Oil Strainer Assembly	8
Assemble Water Pump	8
Install Rear Oil Seal Retainer and Crankshaft Rear Oil Seal	8
Install Oil Pan Gasket	8
Install Oil Pan	9
Assemble Cylinder Head	9
Install Valve Seat Ring	9
Install Valve Guide	9
Press-Fit of Bowl Plug	9
Install Intake and Exhaust Valve	9

Install Valve Spring Lower Seat	10
Install Valve Oil Seal	10
Install Valve Spring, Valve Spring Upper Seat and Valve Cotter	10
Test Air-Leakage of Valve	10
Install Camshaft Assembly and Camshaft Bearing Cap	10
Install Camshaft Oil Seal	11
Press-Fit of Spark Plug Socket	11
Install Rocker Shaft with Rocker Assembly	11
Install Studs on Intake Side	12
Install Studs on Exhaust Side	12
Install Cylinder Head Assembly and Cylinder Head Gasket	12
Install Crankcase Bleed Pipe Joint	13
Install Oil Filter Assembly and Adapter	13
Install Oil Level Indicator	13
Install Timing Train	13
Install Timing Belt and Tensioner	13
Re-Tighten Cylinder Head Rocker Shaft Bolts	14
Install Round Rubber Plug and Valve Cover Assembly	14
Install Semi-Round Rubber Plug	14
Install Valve Cover Assembly	14
Install Camshaft Position Sensor	15
Assemble and Install Thermostat Seat with Thermostat Assembly	15
Install Timing Cover with Gasket Assembly	16
Install Intake Manifold Assembly	16
Install Exhaust Manifold Assembly	16
Fix Oil Level Indicator Tube	16
Install Crankcase Ventilation System	16
Install Spark Plugs, Ignition Coils and Ignition Cable	17

Section 4 Specifications

Major Dimensional Tolerances and Fits of Engine ..	19
Mounting Torques for Major Bolts on Engine	24
Parameters to be Measured during Assembly	26
Oil Sites during Engine Assembly	27
Rubber Coating Positions during Engine Assembly	27

This page intentionally left blank.

Section 1 Introduction and Safety

Introduction

Read this manual thoroughly and understand all of the instructions, cautions, and warnings before using this equipment. If any section of the manual is not understood, contact your nearest Independent Authorized Service Dealer, or contact Generac Customer Service at 1-888-GENERAC (1-888-436-3722), or www.generac.com with any questions or concerns.

The owner is responsible for proper maintenance and safe use of the equipment. Before operating on or servicing this engine:

- Study all warnings in this manual and on the product carefully.
- Become familiar with this manual and the unit before use.
- Refer to the Assembly section of the manual for instructions on final assembly procedures. Follow the instructions completely.

Save these instructions for future reference. ALWAYS supply this manual to any individual that will use this machine.

THE INFORMATION CONTAINED HEREIN WAS BASED ON MACHINES IN PRODUCTION AT THE TIME OF PUBLICATION. GENERAC RESERVES THE RIGHT TO MODIFY THIS MANUAL AT ANY TIME.

Safety Rules

The manufacturer cannot anticipate every possible circumstance that might involve a hazard. The warnings in this manual, and on tags and decals affixed to the unit are, therefore, not all inclusive. If using a procedure, work method or operating technique that the manufacturer does not specifically recommend, verify that it is safe for others. Also make sure the procedure, work method or operating technique utilized does not render the equipment unsafe.

Throughout this publication, and on tags and decals affixed to the engine, DANGER, WARNING, CAUTION and NOTE blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Their definitions are as follows:

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(000001)

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)

CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety warnings cannot eliminate the hazards that they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

Safety Symbols and Meanings



⚠ DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury. (000103)



⚠ DANGER

Electrocution. High voltage is present at transfer switch and terminals. Contact with live terminals will result in death or serious injury. (000129)



⚠ DANGER

Electrocution. Water contact with a power source, if not avoided, will result in death or serious injury. (000104)



⚠ DANGER

Explosion and Fire. Fuel and vapors are extremely flammable and explosive. Store fuel in a well ventilated area. Keep fire and spark away. Failure to do so will result in death or serious injury. (000143)



⚠ WARNING

Risk of Fire. Hot surfaces could ignite combustibles, resulting in fire. Fire could result in death or serious injury. (000110)



⚠ WARNING

Risk of Falling. Do not use this machine or any components on elevated surfaces. Doing so can result in a fall, serious injury, or death. (000114)

⚠ WARNING

Equipment and property damage. Do not alter construction of, installation, or block ventilation for generator. Failure to do so could result in unsafe operation or damage to the generator. (000146)



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury. (000111)



⚠ WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury. (000115)

⚠ WARNING

Accidental start-up. Disconnect spark plug wires when working on unit. Failure to do so could result in death or serious injury. (000141)



⚠ WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury. (000100a)

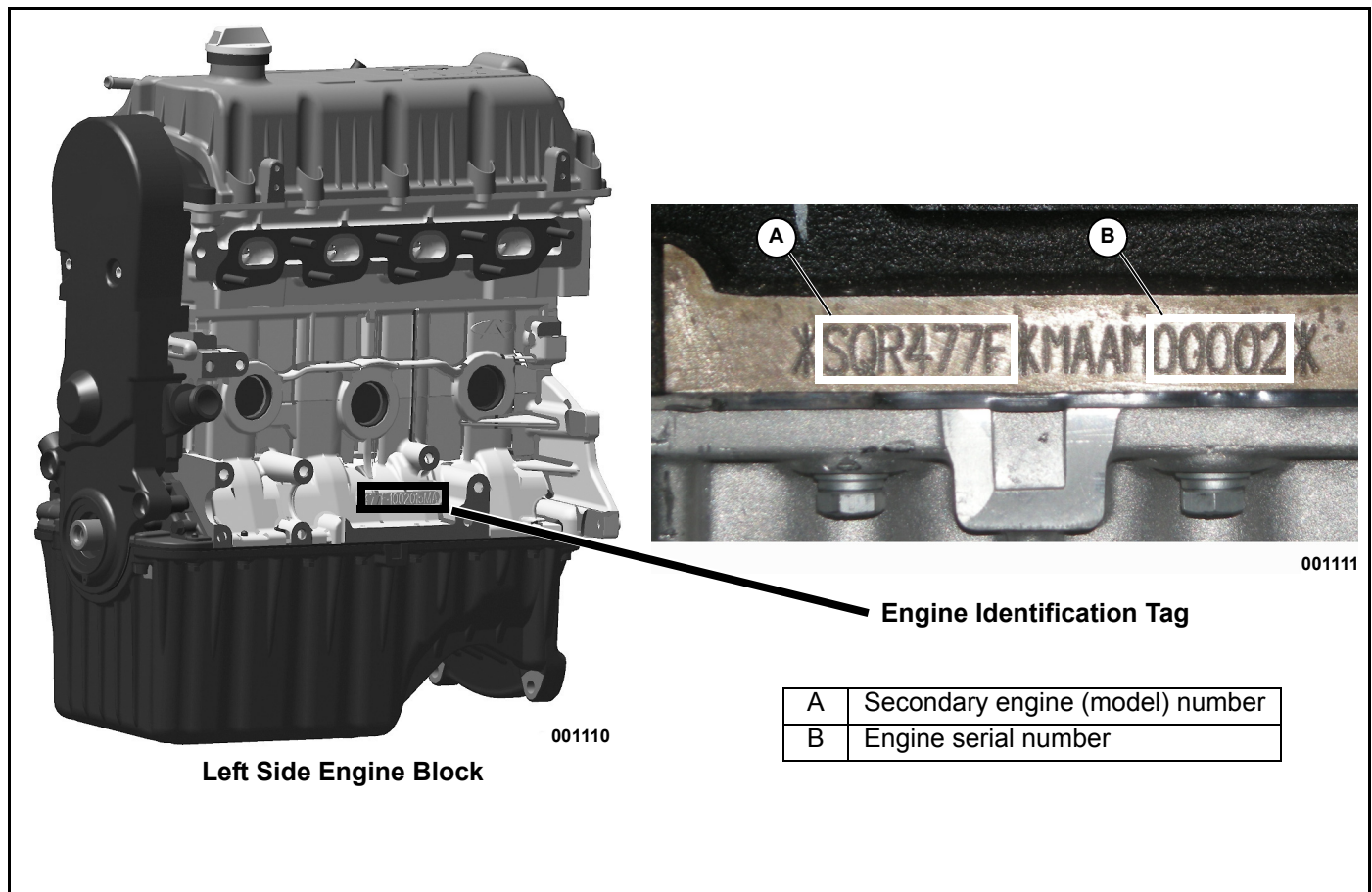
Section 2 General Information

Engine Information

ENGINE SIZE:	1.5 Liter	FUEL:	NG or LPV	Generac (G11)	ENGINE P/N:	0J1556
--------------	-----------	-------	-----------	---------------	-------------	--------

Engine Identification

The engine identification tag is located on the left side of the engine block. The left and right sides are determined by viewing the engine from the flywheel end.



Parts Information

For replacement parts relating to this engine, refer to P/N 0J1556PMNL.

Specifications

Make	Generac
Model	In Line
Cylinders	4
Displacement	1.58 qt (1.497 L)
Firing Order	1-3-4-2
Bore and Stroke	3.05 in x 3.13 in (77.4 mm x 79.5 mm)
Compression Ratio	11:1
Air Intake System	Naturally Aspirated
Spark Plug Gap	0.035 in (0.9 mm)
Direction of Rotation	CCW (from flywheel side)
Dimensions L x W x H	24.8 in x 25 in x 25.6 in (630 mm x 635 mm x 650 mm)
Dry Weight	Net 243 lbs (110 kg)
Oil Capacity	Approximately 4.12 qt (3.9 L) + or - 1 qt depending on oil filter type
Coolant Capacity	Approximately 2 gal (9 L) + or - depending on cooling package being used

Section 3 Assembly and Installation

Assemble Cylinder Block

Grouping and Mark Positions of Cylinder Bores

Under regular production conditions, the diameter tolerance of cylinder block is 0.002 in (0.04 mm) and the cylinder bores fall into four groups. In special cases, increasing cylinder bore diameter is permissible. The cylinder bore diameter tolerance for production and repair is 0.001 in (0.03 mm) and the cylinder bores fall into three groups. When a cylinder bore has been reworked to repair dimension, the other three cylinder bores should be machined to the dimensions for production and repair. See table below for grouped dimensions of cylinder bore diameters:

Group Number	Cylinder Bore Diameter
1	3.05 in (77.36–77.37 mm)
2	3.05 in (77.37–77.38 mm)
3	3.05 in (77.38–77.39 mm)
4	3.05 in (77.39–77.40 mm)
A	3.06 in (77.63–77.64 mm)
B	3.06 in (77.64–77.65 mm)
C	3.06 in (77.65–77.66 mm)

As shown in [Figure 3-1](#), the grouping marks of cylinder bores are stamped on the small rough reference plane on back face of cylinder block on intake side. Grade numbers of bores of cylinders (No. 4, No. 3, No. 2 and No. 1) are stamped from left to right in turn.

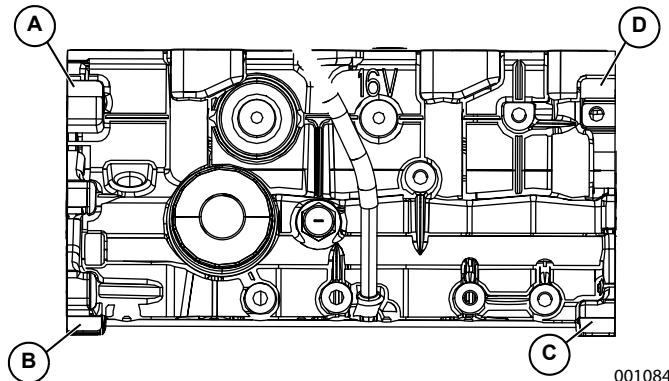


Figure 3-1. Grouping Marks on Cylinder Block

Install Main Bearing Cap

The numbers 1, 2, 3, 4 or 5 and an arrow are cast on each main bearing cap. Install starting from front sequentially. The arrow points to front end face of cylinder block. Width of 1, 2, 4 and 5 main bearing caps are same but No. 3 main bearing cap is wider than the others. Right lower edge of No. 3 main bearing cap requires chamfer. Main bearing caps are positioned by rabbets. Dimensions of a rabbet are not centered so as to prevent installation in opposite direction.

Main Bearing Cap Dimensions	
Long Side	2.27 in (57.7 mm)
Short Side	2.18 in (55.3 mm)

Main bearing cap rabbet and cylinder block main bearing cap rabbet are in interference fit with a magnitude of interference of 0.001–0.006 in (0.025–0.145 mm).

1. Wipe up the mating faces on main bearing caps and cylinder block with clean no-woven fabric or silk cloth before installation.
2. After assembly, side planes of No.1 and No.5 main bearing caps should be flush with front and rear end faces of cylinder block or below them.

Install Main Bearing Cap Bolts

There are nine main bearing cap bolts and one main bearing cap stud. For a transverse engine, the stud is installed in LH side threaded hole (when viewed from front) on No. 2 main bearing cap for mounting oil strainer bracket. For a longitudinal engine, it is installed in LH side threaded hole (when viewed from front) on No. 4 main bearing cap.

1. Apply engine oil to mating faces at heads of main bearing cap bolts.
2. Hand-tighten main bearing cap bolts and main bearing cap stud first, and then tighten them to 797–885 in-lbs (90–100 Nm).

NOTE: If any bolt cannot be screwed in by hand, back it out to locate the cause, replace with a new bolt, or check the bolt hole.

3. After machining of main bearing holes, clean and blow-dry cylinder block thoroughly.

Under regular production conditions, a main bearing hole has only one dimension and will not be grouped. The dimension can be increased by 0.02 in (0.40 mm) for production and repair purposes. The five main bearing holes should be increased at the same time. Stamp size marking of main bearing hole at C in [Figure 3-1](#) to show distinction (increase by 0.02 in (0.40 mm) after dimensioning).

Press-Fit of Bowl Plug

1. Apply Loctite™ 648 sealant around circumference of bowl plug.
2. Drive bowl plug into corresponding hole on cylinder block with a special tool until bowl plug is 0.00–0.02 in (0–0.50 mm) below cylinder block.

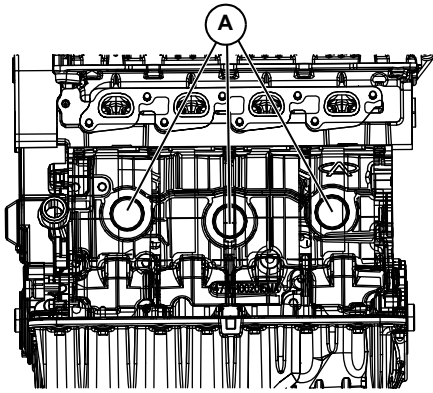


Figure 3-2. Press-Fit of Bowl Plug

001085

Install Inner Hexagon Tapered Plug

Clean up oil passage and apply Loctite™ 243 sealant to the inner hexagon tapered plug with a tightening torque of 177–221 in-lbs (20–25Nm).

Assemble Piston

During assembly, piston skirts are divided into four groups according to piston to cylinder clearance:

Group	Diameter at large point on piston	Bore Diameter	Piston to Cylinder Clearance Before Graphite Treatment
1	3.044–3.045 in (77.325–77.335 mm)	3.05 in (77.36–77.37 mm)	0.001–0.002 in (0.025–0.045 mm)
2	3.045–3.045 in (77.335–77.345 mm)	3.05 in (77.37–77.38 mm)	
3	3.045 in (77.345–77.355 mm)	3.05 in (77.38–77.39 mm)	
4	3.045–3.046 in (77.355–77.365 mm)	3.05 in (77.39–77.40 mm)	

Piston pins holes fall into three groups which are fitted with piston pins in corresponding groups:

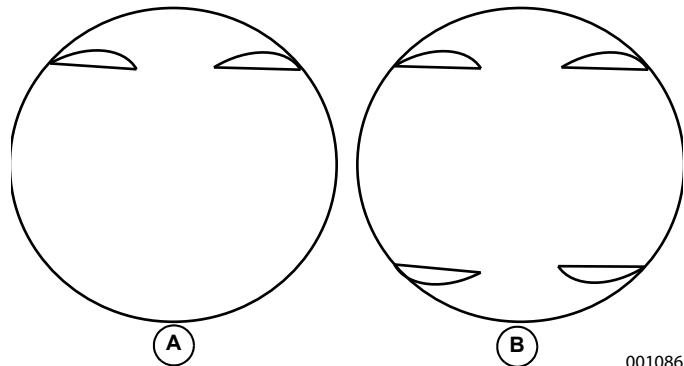
Group	Grouping of Piston Pin Outer Diameters	Inner Diameter of Piston Pin Hole	Mark for Piston Pin Group
1	0.812 in (20.622–20.625 mm)	0.812–0.813 in (20.635–20.638 mm)	White W
2	0.812 in (20.625–20.628 mm)	0.813 in (20.638–20.641 mm)	Red R

3	0.812 in (20.628–20.631 mm)	0.813 in (20.641–20.644 mm)	Blue L
---	-----------------------------	-----------------------------	--------

- Before assembling a piston pin, spray a little engine oil on the circumference and into corresponding piston pin hole.
- Heat up the connecting rod small end to 266–392°F (130–200°C) first, and then press-fit the piston pin into the connecting rod small end hole and piston pin hole.
- After the piston pin is fitted into the piston pin hole, one of the ends should go beyond 0.089–0.125 in (2.268–3.168 mm) from outer edge of the seat with the other end in the pin hole.

NOTE: When assembling, remember to let the F mark on the connecting rod I-beam and the forward mark (arrow) on the piston head face the same direction.

- A piston and connecting rod assembly is formed. Pistons on 477F (A) differ from those on 477FB (B). See [Figure 3-3](#) for the comparison between them.



001086

Figure 3-3. Piston Ratio

Assemble Connecting Rod

Mass of connecting rod assembly (excluding bearing shells): 19.0–19.6 oz (550±6.5g). Weights of the connecting rods on the same unit should be in the same weight group. Dimensions of connecting rod small end holes are not grouped while those of piston pins are grouped. Connecting rod small end hole and corresponding piston pin are in interference fit with a magnitude of interference of 0.000–0.002 in (0.012–0.041 mm). A connecting rod body and corresponding connecting rod cap are positioned by a dowel pin. The dowel pin and the hole on the connecting rod body are in interference fit with a magnitude of interference of 0.00–0.01 in (0.11–0.29 mm); the dowel pin and the hole on the connecting rod cap are in interference fit with a magnitude of interference of 0.00–0.01 in (0.11–0.29 mm).

- Spray engine oil on mating faces of dowel pin holes and dowel pin.

2. After dowel pin is fitted properly, spray engine oil on the connecting rod body, cap and threaded mating faces.
3. Screw on a connecting rod nut by hand and tighten in two steps with a torque wrench:
 - 150–204 in-lbs (20±3 Nm)
 - 266–318 in-lbs (33±3 Nm)

Install Piston Ring

The top piston ring is rectangular, the second gas ring is tapered, and the oil ring is a compound steel band ring.

1. Spray engine oil on piston ring grooves and pistons.

NOTE: Leave the face having words on the top ring upwards and the outer notch on the second ring downwards.

2. When installing oil ring, fit expanders into corresponding oil grooves first.
3. Install the two scrapers. Stagger gaps of the two oil ring scrapers and laps of expanders for 90°, and let the upper and lower scrapers form a 180° angle.
4. Install second gas ring.
5. Install top ring. Stagger the two gas rings and the upper scraper gap for 120°. When installing a ring, use a special tapered sleeve with a maximum diameter 0.04 in (1 mm) larger than diameter of corresponding piston head, so that the piston ring can rotate freely without seizure. The clearances are as follows:

Side Clearance of Piston Ring	
1st Ring Groove	0.00 in (0.04–0.08 mm)
2nd Ring Groove	0.00 in (0.03–0.07 mm)
Oil Ring Groove	0.00–0.01 in (0.03–0.16 mm)
Gap Clearance of Piston Ring	
1st Ring Groove	0.01 in (0.20–0.35 mm)
2nd Ring Groove	0.02 in (0.40–0.55 mm)
Oil Ring Groove	0.01–0.02 in (0.20–0.60 mm)

Assemble Oil Baffle

Install an oil baffle assembly to the right side of tail of cylinder block (when viewed from front). The tension spring of the oil baffle assembly should be below the oil pan flange on cylinder block.

Install Main Bearing Shell

As required by after service, three types of main bearing shells for repair purpose are available:

1. One is for the main bearing hole oversized by 0.02 in (0.4 mm) and has a black mark in addition to “0.4” stamped on the steel back.

2. One is for the crankshaft journal undersized by 0.01 in (0.25 mm) and has a green mark in addition to the mark stamped on the steel back.
3. One is used to correct bearing clearance when former bearing shell generates too much noise, and has a yellow mark in addition to “0.02” stamped on the steel back.

Install Connecting Rod Bearing Shell

Two types of connecting rod bearing shells for repair purpose are available:

1. One is for the crankshaft journal undersized by 0.01 in (0.25 mm) and has a green mark in addition to “0.25” stamped on the steel back.
2. One is used to correct bearing clearance when former bearing shell generates too loud noise and has a yellow mark in addition to “0.025” stamped on the steel back.

When assembling, align oil grooves and oil holes on upper main bearing shell with the oil holes on cylinder block, and verify the following:

- There are no oil holes on lower main bearing shell.
- Upper and lower connecting rod bearing shells can be interchanged.
- Upper and lower connecting rod bearing shells have no oil grooves but each has an oil hole.
- Use a set of connecting rod bearing shells and main bearing shells from the same supplier.
- Before installation, apply engine oil on connecting rod bearing shells and inner surfaces of main bearing shells.
- Keep back surfaces of bearing shells clean.

Install Crankshaft, Crankshaft Woodruff Key and Crankshaft Thrust Plate

Crankshaft key groove and crankshaft woodruff key are in transition fit in width with a magnitude of interference of 0–0.002 in (0–0.051 mm). After a woodruff key is fitted in, protrusion height should be 0.055–0.0685 in (1.392–1.739 mm).

Install two crankshaft thrust plates on front and rear thrust surfaces of the cylinder block.

1. Apply engine oil to each thrust plate. The side without oil groove should face cylinder block and the side with oil groove should face crankshaft.
2. Apply engine oil on main journal, connecting rod journals and thrust surfaces.

Fit clearance of connecting rod bearing	0.000–0.002 in (0.006–0.06 mm)
Fit clearance of main bearing	0.000–0.002 in (0.011–0.058 mm)

3. Measure axial clearance of crankshaft. Clearance should be within 0.004–0.012 in (0.093–0.303 mm). Measure rotating torque of crankshaft. Rotating

torque should be less than 142 **in-lbs** (16 Nm) after crankshaft rotates a maximum of six turns after piston and connecting rod assembly is installed.

4. After installing cylinder head without spark plugs, rotating torque of crankshaft should be less than 274 **in-lbs** (31 Nm) after crankshaft rotates a maximum of six turns.

Install Oil Pump, Crankshaft Front Oil Seal and Oil Strainer

Assemble Front Oil Seal

1. Install oil pump onto press-fit fixture. The face with the part number should be outside.
2. Press crankshaft front oil seal in place. Check oil seal edge for damage and defects. Magnitude of interference between front oil seal and oil pump housing: 0.01–0.02 in (0.26–0.50 mm).

Install Oil Pump with Front Oil Seal Assembly

1. Install oil pump onto cylinder block.
2. Hand-tighten the two bolts.
3. Hand-tighten the other four bolts after the process guide pin is removed.
4. Tighten oil pump bolts to 71–97 **in-lbs** (8–11 Nm).
5. Measure flatness between oil pump bottom plane and oil pan flange on cylinder block, which should be ± 0.01 in (± 0.26 mm).

Install Oil Strainer Assembly

1. Install oil strainer gasket onto oil pump inlet flange.
2. Install oil strainer assembly.
3. Hand-tighten the two oil strainer flange washer bolts.
4. Tighten an oil strainer bracket onto cylinder block using a bolt. Put bracket over the stud on the left side of No. 2 main bearing cap.
5. Hand-tighten nut after installing a plain washer.
6. Tighten oil strainer flange bolts to 71–106 **in-lbs** (8–12 Nm).
7. Tighten bracket bolt to 150–204 **in-lbs** (17–23 Nm).
8. Oil strainer assembly should be installed onto cylinder block and oil pump freely. If any stress is found during assembly, replace it.

Assemble Water Pump

1. Install two process guide pins on front face of cylinder block.

2. Fit a water pump gasket through the process guide pins and install water pump assembly onto cylinder block.
3. Remove process guide pins.
4. Hand-tighten the four water pump bolts and tighten to 71–97 **in-lbs** (8–11 Nm).

NOTE: Water pump gear and pump impeller should rotate freely.

Install Rear Oil Seal Retainer and Crankshaft Rear Oil Seal

1. Before assembling rear oil seal retainer with rear oil seal assembly, apply engine oil on rear oil seal edge or crankshaft rear oil seal journal.
2. Stretch edge with a special tool, keeping the oil seal edge clear of the tool to avoid damage.
3. Install two process guide pins on rear end face of cylinder block.
4. Fit a rear oil seal retainer onto cylinder block through the process guide pins.
5. Install rear oil seal retainer with rear oil seal assembly onto cylinder block using positioner.
6. Hand-tighten two bolts.
7. Remove the process guide pins and tighten the other two bolts.
8. Tighten the four bolts to 71–97 **in-lbs** (8–11 Nm).
9. Measure flatness between rear oil seal retainer bottom plane and oil pan flange on cylinder block, which should be within ± 0.01 in (± 0.26 mm).

Install Oil Pan Gasket

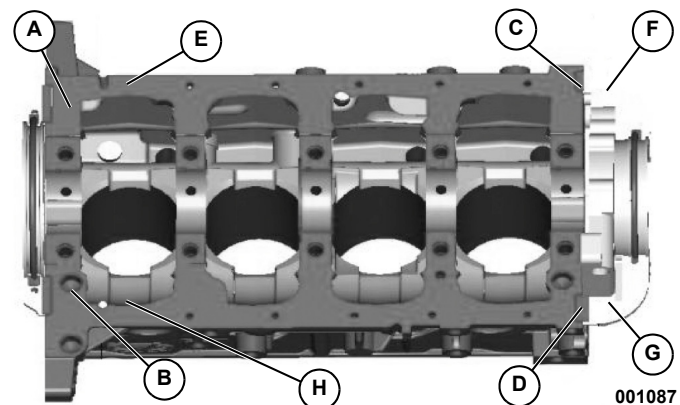


Figure 3-4. Sealant Pre-Coating Positions Before Installation of Oil Pan Gasket

1. See [Figure 3-4](#). Coat locations A–D with $\varnothing 0.08$ – 0.12 in ($\varnothing 2$ – 3 mm) Loctite™ 5699 sealant evenly (pay extra care to the places where oil pump and rear oil seal retainer arcs transit to planes).
2. Insert tooling dowel pins at E and F to fit on oil pan gasket and verify that the gasket contacts rear oil seal retainer and oil pump, and fits evenly.

Install Oil Pan

1. Coat gasket with Loctite™ 5699 sealant and install oil pan through the dowel pins.

Installation of oil pan bolts (See [Figure 3-5](#)):

2. Pre-tighten the socket head cap screw units at places A–D by hand.
3. Pre-tighten the hexagon bolts at places 1–10.
4. Pull out the process guide pins at places 11–14, and then pre-tighten the other four bolts.
5. When pre-tightening, each bolt should be screwed in by 1–2 threads.

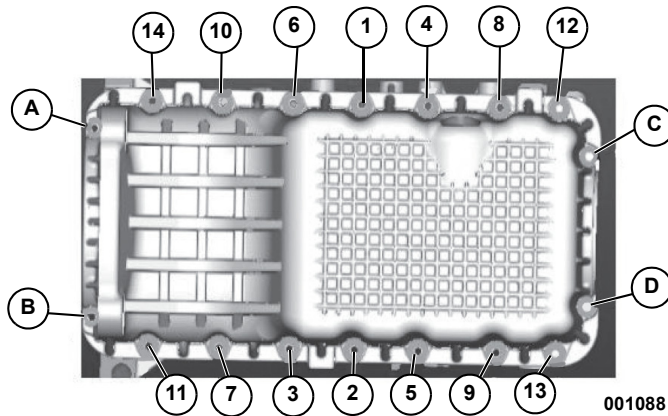


Figure 3-5. Pan Oil Tightening

Methods for tightening oil pan bolts (See [Figure 3-5](#)):

1. Multi-axis tightening: tighten 18 bolts to 53–71 **in-lbs** (7 ± 1 Nm) simultaneously.
2. Tightening using single socket wrench:
 - Tighten bolts (1–14) to 53–71 **in-lbs** (7 ± 1 Nm) in the order shown.
 - Tighten the four inner hexagon bolts (A–D) to 53–71 **in-lbs** (7 ± 1 Nm) in the order shown.

Assemble Cylinder Head

Install Valve Seat Ring

Intake and exhaust valve seat rings and corresponding cylinder head holes are in interference fit with a magnitude of interference of 0.003–0.004 in (0.078–0.11 mm).

1. Before pressing valve seat rings, wash cylinder head and blow-clean valve seat ring holes with high pressure air.
2. Soak valve seat rings in a liquid nitrogen tank for at least ten minutes.
3. Remove valve seat rings from liquid nitrogen tank and press them in cylinder head using a special tool.

NOTE: The valve seat rings must be pressed into place without clearance between bottom of each seat ring hole and corresponding valve seat ring.

4. Check seating condition of each valve seat ring.

Install Valve Guide

Valve guide and valve guide hole on cylinder head are in interference fit with a magnitude of interference of 0.001–0.002 in (0.022–0.051 mm).

1. Soak valve guides in a liquid nitrogen tank for at least ten minutes.
2. Take valve guides out of the liquid nitrogen tank and press them into cylinder head.
3. After being pressed in, height of upper end face of each valve guide should meet the requirements of cylinder head assembly.

Press-Fit of Bowl Plug

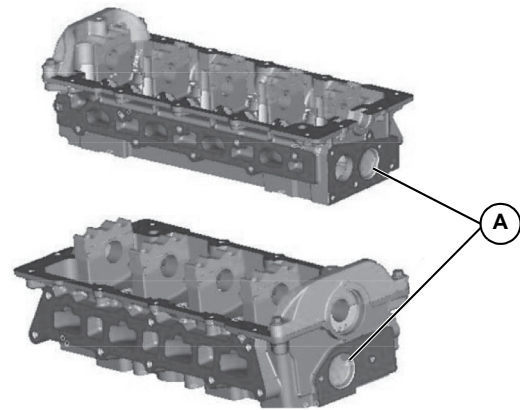


Figure 3-6. Press-Fitting Bowl Plugs in Cylinder Head

A	Water jacket cleaning holes
---	-----------------------------

4. After machining and washing cylinder head, press bowl plugs in to seal them.
5. Apply Loctite™ 648 sealant around circumferences of bowl plugs and press them into the mounting holes on cylinder head using a special tool.

NOTE: After being pressed in, bowl plugs should be 0.00–0.02 in (0–0.5 mm) below chamfer surfaces of corresponding holes on cylinder head.

Install Intake and Exhaust Valve

There are two intake valves and two exhaust valves on each cylinder. Among these valves, area of intake valve head is larger than that of exhaust valve head.

1. Remove No. 1 bearing cap and No. 2–No. 5 bearing caps and put them in order.
2. Verify that burrs produced during machining have been removed completely. Check cylinder head and camshaft cover for damage.
3. Clean all oil passages and inner chambers on cylinder head. Verify all oil passages are unobstructed and there are no burrs at joints of oil passages.

Install Valve Spring Lower Seat

- Slide valve spring lower seat (bottom plane faces cylinder head) into valve guide. Valve spring lower seat bottom plane should stick to the machined surface on valve spring lower seat on cylinder head.

Install Valve Oil Seal

- Apply lubricating oil on lip of each valve oil seal and install valve oil seal assemblies onto corresponding valve guides.
- Tap valve guides using pressing sleeve and hammer to fit oil seals in place. Verify oil seals and valve guides fit in position.
- Tilt cylinder head to wipe up intake and exhaust valves and valve seat rings using fabric or silk cloth.
- Apply engine oil to valve stems.
- Fit intake and exhaust valves into cylinder head.

NOTE: No valve oil seals should move when sliding valves up and down.

Fit clearance between intake valve stem and valve guide hole	0.000–0.002 in (0.006–0.06 mm)
Fit clearance between exhaust valve stem and valve guide hole	0.000–0.002 in (0.011–0.058 mm)

Install Valve Spring, Valve Spring Upper Seat and Valve Cotter

- Lock cylinder head with valve mechanism fitted in on a workbench.
- Fix valve cotter press-fitting tool onto upper plane of cylinder head.
- Put valve springs and valve spring upper seats in from valve guides in turn.

NOTE: Valve spring is a variable pitch spring of which the end coated with paint must face valve spring base.

- Put pressure head on upper seat, depress valve spring, fit valve cotter into groove using a forceps, and then loosen pressure head to allow cotter to grip valve.
- Beat top of each intake and exhaust valve stem using a copper hammer to seat valve coppers firmly and help seal.

Test Air-Leakage of Valve

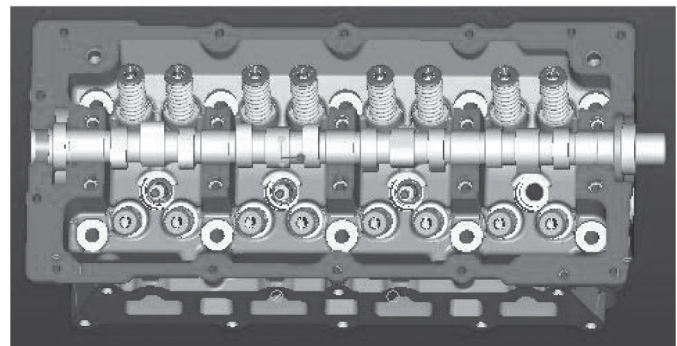
- Fill intake and exhaust passages with test gas. Results should be as follows:
 - Pressure: 7 psi (50 kPa)

- Gas leakage: ≤ 3 oz/min (100ml/min)
- If any valve is does not meet these specifications, replace and retest.

Install Camshaft Assembly and Camshaft Bearing Cap

Camshaft woodruff key and camshaft key groove are in transition fit in width with a fit clearance of -0.002–0.000 in (-0.051–0.004 mm).

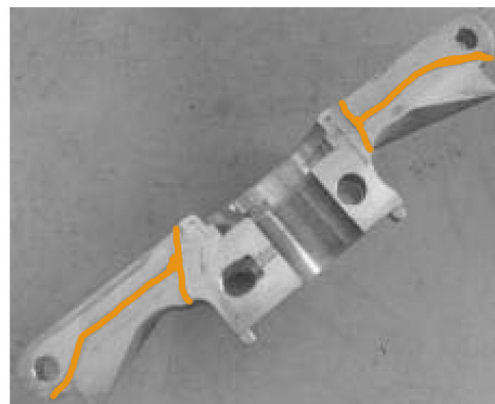
- After a woodruff key is fitted in, protrusion height should be 0.06–0.08 in (1.64–2.11 mm).
- Clean camshaft holes on cylinder head with silk cloth and apply engine oil to them.
- See [Figure 3-7](#). Install camshaft assembly with woodruff key fitted into the bearing saddle holes on cylinder head.



001090

Figure 3-7. Camshaft Assembly

- Clean all camshaft cover holes and apply sealant to mating face of No. 1 bearing cap (Loctite™ 574). Sealant coating diameter should be <0.04 in (1 mm).
- See [Figure 3-8](#) for sealant coating zones.



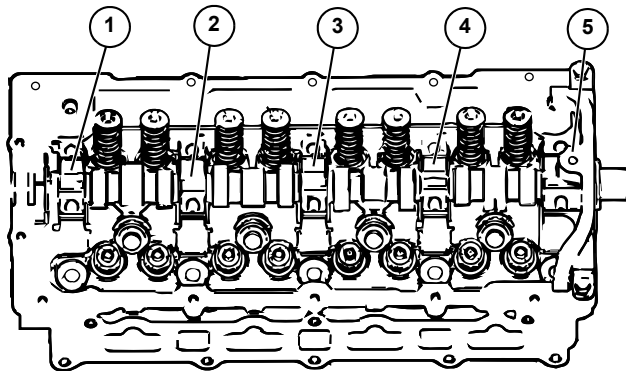
001091

Figure 3-8. Camshaft with Sealant

- Apply engine oil to surfaces of camshaft cams and journals.
- See [Figure 3-9](#). Install No.1–No.5 bearing caps sequentially.

Camshaft cover and dowel pin on cylinder head are in clearance fit with a fit clearance of 0.000–0.001 in (0.005–0.028 mm).

8. Align location hole of camshaft cover with dowel pin, and gently press the cover until attached.



001092

Figure 3-9. Camshaft Bearing Saddles

9. Turn camshaft by hand to rotate smoothly with a rotating torque less than 9 in-lbs (1 Nm).
10. Turn camshaft until key groove faces down.
11. Inspect axial clearance of camshaft after assembly with a dial gauge. When pushing camshaft back and forth, the axial clearance should be within 0.004–0.006 in (0.095–0.153 mm).

Install Camshaft Oil Seal

Camshaft journal and oil seal bore are in interference fit with a magnitude of interference of 0.056–0.085 in (1.425–2.15 mm).

Oil seal seat bore and oil seal outer diameter are in interference fit with a magnitude of interference of 0.01–0.02 in (0.17–0.40 mm).

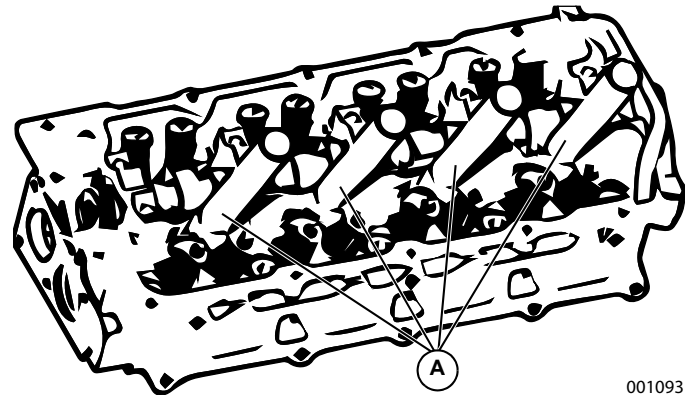
1. Spray-coat a layer of engine oil to inner and outer edges of an oil seal and camshaft oil seal journal.
2. Press the oil seal into cylinder head using a special tool with the part number side facing out.
3. External surface of the oil seal should be flush with corresponding oil seal seat hole on cylinder head (deviation: -0.02–0.00 in (-0.5–0.00 mm)).

Press-Fit of Spark Plug Socket

Spark plug socket and spark plug socket hole on cylinder head are in interference fit with a magnitude of interference of 0.003–0.004 in (0.067–0.101 mm).

1. Apply Loctite™ 648 sealant about 0.8 in (2 cm) from lower end of spark plug socket.
2. See [Figure 3-10](#). Drive socket into the bottom hole using a special tool until it becomes flush with the step surface of the bottom hole.

NOTE: Differentiation between upper and lower ends of spark plug socket is not required.



001093

Figure 3-10. Spark Plug Socket

Install Rocker Shaft with Rocker Assembly

Rocker shaft falls into intake rocker shaft and exhaust rocker shaft. Four grooves are machined on intake rocker (for avoiding spark plug socket).

1. Examine oil holes and oil grooves on each rocker shaft for burrs and fins. Exterior of rocker shaft should be clean and free of residue and foreign material.

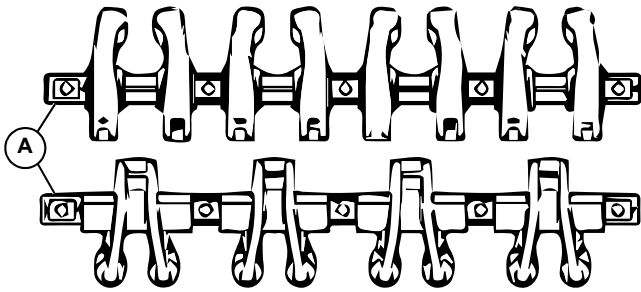


001094

Figure 3-11. Intake and Exhaust Rocker Shafts

2. Inspect ball heads and retainers of intake and exhaust rockers for damage.
3. Apply engine oil to rocker shafts.
4. Install rockers onto corresponding rocker shafts as shown in [Figure 3-12](#).
5. Assemble intake and exhaust rocker shafts. The end with the larger chamfer (A) should face No. 1 bearing cap.

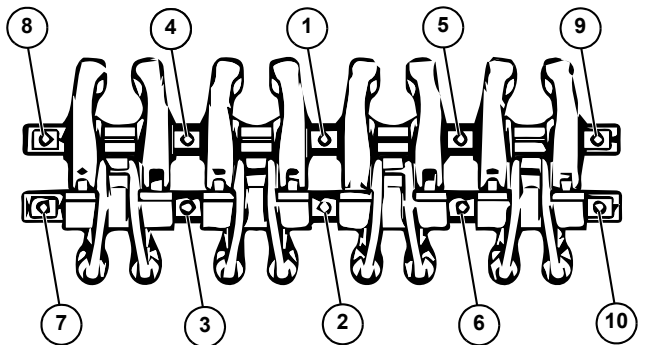
IMPORTANT NOTE: Installation in opposite direction will fail.



001095

Figure 3-12. Rocker Shaft with Rocker

6. Verify the camshaft key groove faces down
7. Install the rocker shaft with rocker assembly in intake side.
8. Adjust positions of all rockers until rocker shaft falls on bearing saddle bracket.
9. Check all intake rocker shafts for axial seizure.
10. Press on intake rocker stopper between intake rocker shafts.
11. Install rocker shaft with rocker assembly on exhaust side.
12. Adjust positions of all rockers until rocker shaft falls on bearing saddle bracket.
13. Verify all rockers are in their installation positions.
14. Install rocker shaft bolts.
15. Tighten bolts as follows:
 - Tighten to 119–146 **in-lbs** (15 ± 1.5 Nm) in the numerical order shown in **Figure 3-13**.
 - Tighten bolts for the second time after cylinder head assembly is fitted onto cylinder block. Pre-tighten to 252–279 **in-lbs** (30 ± 1.5 Nm) in the numerical order shown in **Figure 3-13**.
16. After rocker shaft bolts are pre-tightened, tighten the bolts on both sides of No. 1 bearing cap to 71–97 **in-lbs** (9.5 ± 1 Nm).

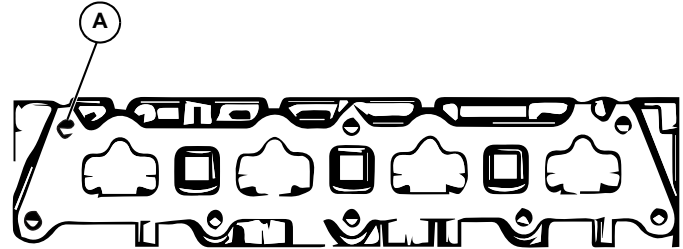


001096

Figure 3-13. Rocker Shaft with Rocker Assembly

Install Studs on Intake Side

1. Apply sealant on heads of the stud bolts on intake side (sealant type: Loctite™ 262 or Loctite™ 271).
2. Screw in bolts on intake side by 2–3 threads by hand.
3. See **Figure 3-14**. Install a stud (A) into the bolt hole at top left corner on intake side.
4. Install seven equal-length studs on the exhaust side and tighten to 71–106 **in-lbs** (10 ± 2 Nm).



001097

Figure 3-14. Install Studs on Intake Side

Install Studs on Exhaust Side

1. Apply sealant to heads of stud bolts on exhaust side (sealant type: Loctite™ 262 or Loctite™ 271).
2. Screw in bolts on exhaust side by 2–3 threads by hand.
3. Install eight equal-length studs on the exhaust side and tighten to 71–106 **in-lbs** (10 ± 2 Nm).

Install Cylinder Head Assembly and Cylinder Head Gasket

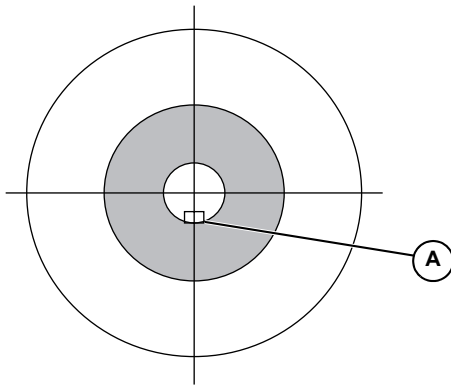
1. Install two dowel pins on upper plane of cylinder block. Cylinder head dowel pins (spring dowel pins) are on No. 2 and No. 4 threaded holes at LH side (when viewed from front) of cylinder block.

Magnitude of interference between dowel pin and cylinder block: 0.002–0.008 in (0.053–0.203 mm). Dowel pin and cylinder head are in transition fit with a clearance of -0.001–0.003 (-0.016–0.084 mm).

2. Verify cylinder head gasket is smooth and clean without bumps and scratches.
3. Let the side with the part number stamped on the cylinder head gasket face up.
4. Fit cylinder head gasket onto cylinder block plane through the dowel pins.
5. Wipe up mating surfaces between cylinder head, combustion chambers, top plane and threaded holes on cylinder block.

IMPORTANT NOTE: Verify no oil has accumulated at the bottom of any threaded hole on cylinder block.

6. Keep camshaft key groove at the position shown (A) in **Figure 3-15** (when viewed from front end of camshaft).



001098

Figure 3-15. Position of Camshaft Key Groove

Installation of cylinder head bolt:

1. Leave the cylinder head bolt washer face having chamfer upwards and the comparatively flat face towards cylinder head.
2. After the washer is installed on the cylinder head bolt, put the bearing face at bolt head and the threaded portion into the cylinder head bolt hole and tighten the bolt by 3–4 threads.
3. Tighten cylinder head bolts as follows:
 - Tighten to **177–221 in-lbs** (20–25 Nm) in the numerical order shown in **Figure 3-13**.
 - Tighten to **354–398 in-lbs** (40–45 Nm) in the numerical order shown in **Figure 3-13**.
 - Turn 90° in tightening order.
 - Turn another 90° in tightening order

NOTE: Bolts assembled using this method cannot be reused.

4. In case of any error during tightening, repair as follows:
 - Parameters of one or two cylinder head bolts that are not adjacent are incorrect—replace the cylinder head bolt(s).
 - Parameters of two adjacent or three and more cylinder head bolts are incorrect—loosen all cylinder head bolts and replace them, replace cylinder head gasket and tighten all cylinder head bolts manually.

NOTE: Replaced cylinder head bolts cannot be reused and should be discarded.

Install Crankcase Bleed Pipe Joint

1. Press crankshaft bleed pipe joint into cylinder block until the locating boss is 0.06 in (1.5 mm) maximum from upper plane of cylinder block bleed pipe hole.

Crankshaft bleed pipe and cylinder block bleed pipe hole are in interference fit with a magnitude of interference of 0.01 in (0.25 mm).

Install Oil Filter Assembly and Adapter

Oil filter adapter uses inch screw external thread.

1. Screw the adapter into cylinder bore by hand and then tighten to **124–186 in-lbs** (14–21 Nm) using a special tool.
2. Lubricate oil filter gasket surface with engine oil.
3. Screw oil filter in by hand and tighten to **106–150 in-lbs** (12–17 Nm).

Install Oil Level Indicator

1. Lock the press-fitting tool on the fixture block.
2. Use a hammer to fit the oil level indicator tube into place when the limit boss and cylinder block join.
3. Insert the oil level indicator into tube.

Install Timing Train

1. Turn crankshaft to No. 1 cylinder TDC. The crankshaft woodruff key is at top.
2. Put crankshaft timing gear gasket over crankshaft first during which let the convex face forward.
3. Put crankshaft timing gear over crankshaft.
4. The side with the word “FRONT” should face front and leave TDC mark on the gear upwards.
5. Install camshaft timing gear and check the TDC position on the gear, which should be aligned with the TDC mark on cylinder head.
6. Install gear backing plate and bolts.
7. After screwing in the bolts by hand, tighten them to **797–885 in-lbs** (95± 5 Nm). When tightening, lock up camshaft.

Install Timing Belt and Tensioner

1. Push tensioner to the right (when viewed from front) as far as possible.
2. Hand-tighten the two set bolts and then tighten them to **142–177 in-lbs** (16–20 Nm).
3. When installing the timing belt, make sure that the camshaft is in no-load state and the rocker is in working position but not clamped.
1. Set crankshaft to No. 1 cylinder TDC and verify that the TDC mark on camshaft timing gear aligns with the mark on front end face of cylinder head.
2. Assemble timing belt counterclockwise starting from crankshaft timing gear.

NOTE: During assembly, crankshaft and camshaft must be locked up.

3. Loosen tensioner bolt for 1/2-1 turn and push tensioner towards timing belt.
4. Use an auxiliary fixture to apply 2213–3098 **in-lbs** (300±50 Nm) load to tensioner in horizontal direction.
5. Maintain the load applied.
6. Rotate crankshaft clockwise two turns until it rotates slowly to No.1 cylinder TDC position.

NOTE: Do not rotate crankshaft counterclockwise.

7. Verify that the TDC mark on camshaft timing gear aligns with that on front end of cylinder head
8. Use a tension tester to determine tension of timing belt (the value should be 2213–3098 **in-lbs** (250–350 Nm)).

After adjustment, tension of timing belt (when camshaft is in no-load state) should be 2213–3098 **in-lbs** (250–350 Nm) when measured using a tension tester. The tension tester should be calibrated beforehand. This inspection should be performed immediately after timing belt adjustment (before engine runs again).

Re-Tighten Cylinder Head Rocker Shaft Bolts

1. Re-tighten cylinder head rocker shaft bolts as follows:
2. Tighten to 252–279 **in-lbs** (30±1.5 Nm) in the numerical order shown in **Figure 3-16**.

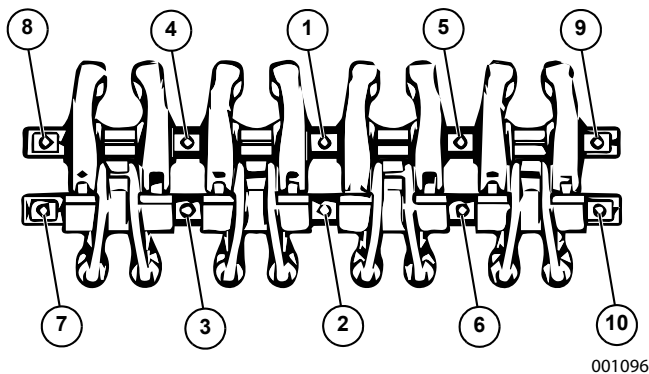


Figure 3-16. Tightening Order of Rocker Shaft Bolts

Install Round Rubber Plug and Valve Cover Assembly

Install Semi-Round Rubber Plug

1. Apply Loctite™ 5910 sealant Ø0.08–0.12 in (Ø2–3mm) to the colored area (A) in **Figure 3-17**.

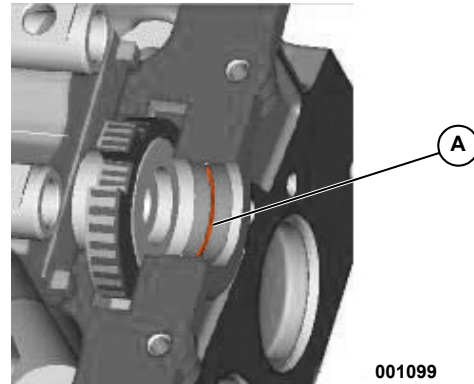


Figure 3-17. Install Rubber Plug (1)

2. See **Figure 3-18**. Install the semi-round rubber plug (A) with limit groove aligned with the TDC on cylinder head.

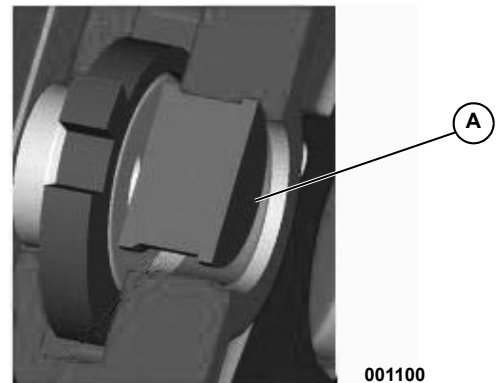
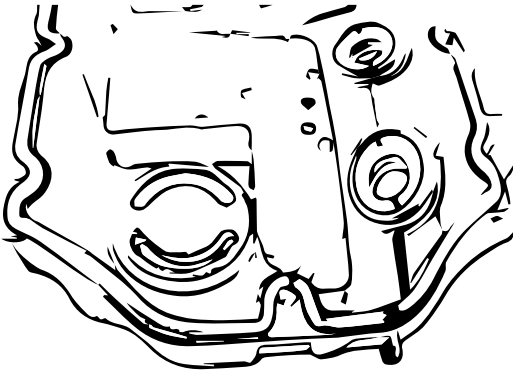


Figure 3-18. Install Rubber Plug (2)

Install Valve Cover Assembly

NOTE: When assembling, handle with care to prevent the gasket from coming off the valve cover. Assemble in place at one time to prevent sealant deflection.

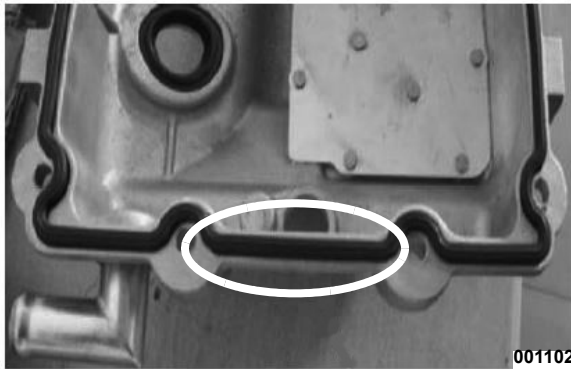
3. Apply sealant to valve cover.
4. Install valve cover gasket into slot.
5. Apply sealant at both the portal shaped line where front end of valve cover joins with bearing cap and the valve cover trough and gasket portion between the two bolts at rear end as shown in **Figure 3-19–Figure 3-21**.
6. Apply a layer of Loctite™ 5910 sealant to corresponding positions on the gasket.



001101

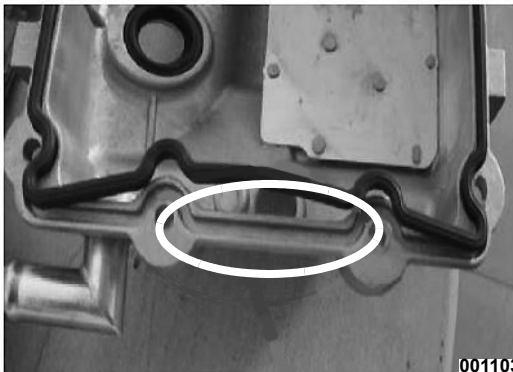
Figure 3-19. Sealant Coating of Valve Cover (1)

7. Install valve cover onto cylinder head after being aligned.
8. Slide spark plug tube through the gasket and press it hard until valve cover and the bolt holes on cylinder head engage and fit well.



001102

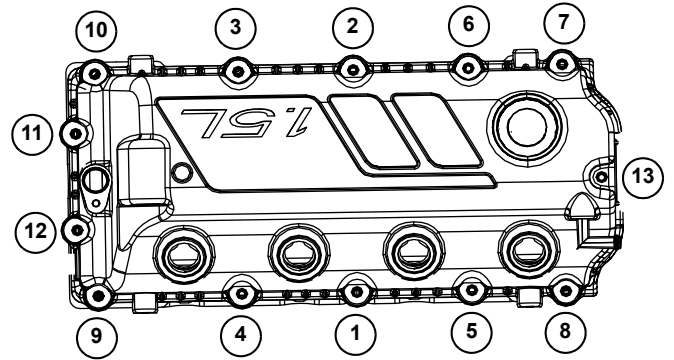
Figure 3-20. Sealant Coating of Valve Cover (2)



001103

Figure 3-21. Sealant Coating of Valve Cover (3)

9. Fit valve cover bolts on valve cover.
10. Pre-tighten bolts to 27–44 **in-lbs** (4 ± 1 Nm) first.
11. Re-tighten bolts to 53–71 **in-lbs** (6–8 Nm).
12. Bolts are used at marks 1–13 in [Figure 3-22](#). See [Figure 3-22](#) for tightening order.



001104

Figure 3-22. Tightening Order of Valve Cover Bolts

Install Camshaft Position Sensor

1. Apply a layer of salve to camshaft position sensor gasket.
2. Put on an O-ring.
3. Install them onto valve cover and tighten to 66–75 **in-lbs** (8 ± 0.5 Nm) using an inner hexagon bolt.
4. Clearance standard between camshaft position sensor and signal wheel of speed sensor: 0.03–0.04 in (0.8–1.0 mm).

Assemble and Install Thermostat Seat with Thermostat Assembly

NOTE: Before installation, apply Loctite™ 243 sealant to coolant temperature sensor.

5. Install thermostat assembly into thermostat seat assembly.
6. Install thermostat seat cover.
7. Install bolts and tighten to 71–97 **in-lbs** (8–11 Nm).

NOTE: Remember to put the protruding point on thermostat gasket into the groove on the hole seat.

8. Tightening torque of coolant temperature sensor: 177–221 **in-lbs** (20–25 Nm).
9. Check the mating face between thermostat seat and cylinder head for bumps, bosses and scratches.
10. Fix thermostat assembly to rear end face of cylinder head using three M6×60 bolts, and then tighten the three bolts to 71–97 **in-lbs** (8–11 Nm).

NOTE: Thermostat seat gasket can only be assembled once and should be replaced at the second assembly.

Install Timing Cover with Gasket Assembly

1. Install upper and lower covers onto front end face of engine as shown in [Figure 3-23](#).
2. Hand-tighten cover bolts by 2-3 screw threads, use a pneumatic wrench to tighten the bolts in place, and then re-tighten them to 71–97 **in-lbs** (8–11 Nm) using a constant torque wrench.

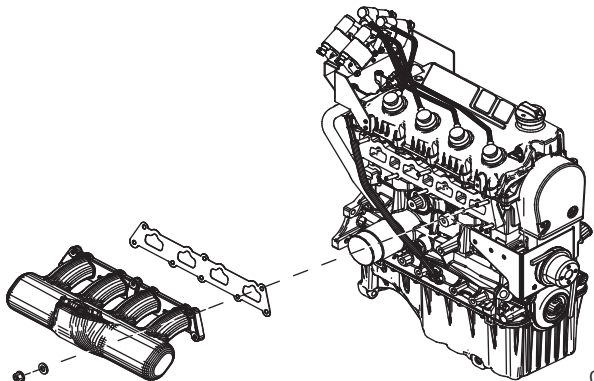


001105

Figure 3-23. Installation of Timing Cover

Install Intake Manifold Assembly

1. See [Figure 3-24](#). Install intake manifold gasket (A) onto intake side of cylinder head through stud bolts (B) on intake side, with the face having words on the gasket facing outside.
2. Install intake pipe (C) onto cylinder head through studs on intake side (slide oil level indicator through intake manifolds of No.3 and No.4 cylinders).
3. Install front hanger onto the stud bolts at front end of cylinder head.
4. Screw five hexagon flange nuts onto corresponding studs by 2-3 threads by hand. Do not tighten the two nuts on left side and the one nut at middle top at this time.

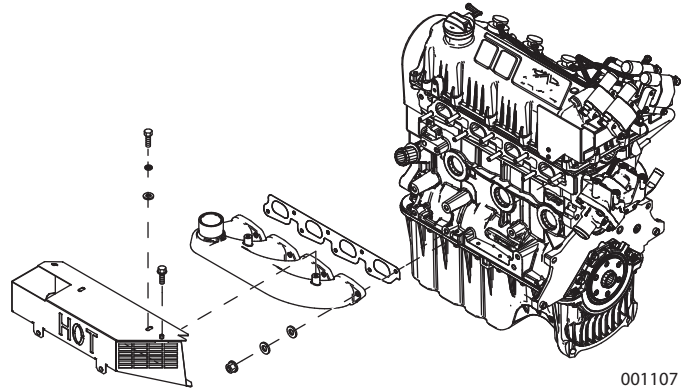


001106

Figure 3-24. Installation of Intake Manifold Assembly

Install Exhaust Manifold Assembly

1. Install exhaust manifold gasket to exhaust side of cylinder head through stud bolts on exhaust side.
2. Install exhaust manifold onto cylinder head.
3. Put a plain washer over each bolt, and then hand-tighten six nuts (do not tighten the two nuts on the right side at this time).



001107

Figure 3-25. Installation of Exhaust Manifold Assembly

Fix Oil Level Indicator Tube

1. See [Figure 3-26](#) for states of the oil level indicator tube bound properly.



001108

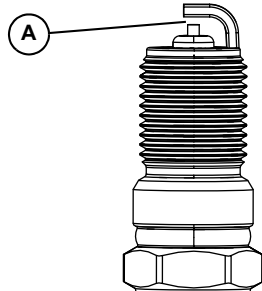
Figure 3-26. Fixing Oil Level Indicator

Install Crankcase Ventilation System

1. Press rubber sleeve of positive crankcase valves (PCV) into valve cover and then drive in PCV valves in turn.
2. Put clips over both ends of hose and lock the ends with clips to corresponding position.

Install Spark Plugs, Ignition Coils and Ignition Cable

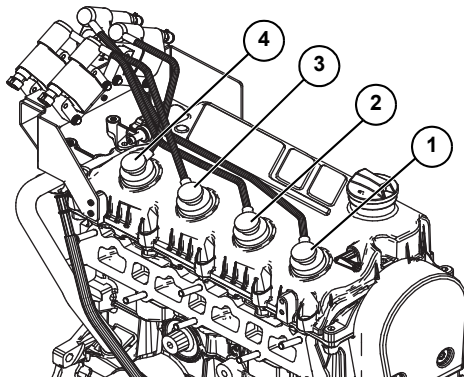
1. Inspect electrode gap with wire feeler gauge and set spark plug gap (A) to 0.035 in (0.9 mm).



001186

Figure 3-27. Spark Plug Gap

2. Apply a light coat of anti-seize to spark plugs and hand thread them into each cylinder. Tighten respectively, to 239–292 in-lbs (30±3 Nm).
3. Install ignition coils onto thermostat seat and tighten fastening bolts (4 x M6 x 50) to 71–97 **in-lbs** (8–11 Nm).
4. Using dielectric grease, connect the spark cables to the appropriate ignition coil and respective spark plug.
5. See [Figure 3-28](#) for cylinder identification.



001109

Figure 3-28. Installation of Ignition Coils, Layout of Ignition Cables and Cylinder Identification

This page intentionally left blank.

Section 4 Specifications

Major Dimensional Tolerances and Fits of Engine

Dimensional Tolerances and Fits (page 1 of 5)

	Name	Dimension and Tolerance	Grouped Dimensions		Fit Clearance
1	Cylinder bore on cylinder block of 477F/477FB engine	—	1	77.36–77.37	0.001–0.002 in (0.025–0.045 mm)
			2	77.37–77.38	
			3	77.38–77.39	
			4	77.39–77.40	
2	Piston skirt on 477F/477FB engine	—	1	77.325–77.335	—
			2	77.335–77.345	
			3	77.345–77.355	
			4	77.355–77.365	
3	Outer diameter of piston pin non 477F/477FB engine	—	White W	20.622–20.625	0.000–0.001 in (0.010–0.016 mm)
			Red R	20.625–20.628	
			Blue L	20.628–20.631	
4	Outer diameter of piston pin on 477F/477FB engine	—	White W	20.622–20.625	0.000–0.002 in (0.012–0.041 mm)
			Red R	20.625–20.628	
			Blue L	20.628–20.631	
5	Diameter of connecting rod small end on 477F/477FB engine	20.6± 0.01	—	—	0.000–0.002 in (0.012–0.041 mm)
	Name	Dimension and Tolerance	Fit Clearance		
6	Width of main bearing saddle on cylinder block	113.00 ± 0.03	-0.001– -0.005 in (-0.025– -0.135 mm)		
	Width of rabbet on main bearing cap	113.055–113.105			
7	Diameter of dowel pin hole on connecting rod body	ϕ 4.0 $\begin{smallmatrix} +0 \\ -0.05 \end{smallmatrix}$	0.00– -0.01 in (-0.11– -0.29 mm)		
	Diameter of spring dowel pin on connecting rod	ϕ 4.11– ϕ 4.24			
	Diameter of dowel pin hole on connecting rod cap	ϕ 4.0 $\begin{smallmatrix} +0 \\ -0.05 \end{smallmatrix}$			
8	Width of connecting rod big end	22.3+0.058	0.011–0.004 in (0.268–0.092 mm)		
	Width of crankshaft connecting rod journal	22.5±0.05			
9	Height of piston 2nd ring groove	1.2 $\begin{smallmatrix} +0.05 \\ +0.03 \end{smallmatrix}$	0.00 in (0.04–0.08 mm)		
	Height of top gas ring	1.2 $\begin{smallmatrix} +0.05 \\ +0.03 \end{smallmatrix}$			
10	Height of piston 2nd ring groove	1.5 $\begin{smallmatrix} +0.04 \\ +0.02 \end{smallmatrix}$	0.00 in (0.03–0.07 mm)		
	Height of 2nd gas ring	1.5 $\begin{smallmatrix} +0.04 \\ +0.02 \end{smallmatrix}$			

Dimensional Tolerances and Fits (page 2 of 5)

	Name	Dimension and Tolerance	Fit Clearance
11	Height of piston oil ring groove	2 $\begin{smallmatrix} +0.03 \\ +0.01 \end{smallmatrix}$	0.00–0.01 in (0.03–0.16 mm)
	Height of steel band compound oil ring scraper	0.45 max	
	Height of steel band compound oil ring expander	1.8 $\begin{smallmatrix} -0.05 \\ -0.15 \end{smallmatrix}$	
12	Diameter of connecting rod bearing hole	ϕ 47.9 $\pm$ 0.01	0.00 in (0.006–0.06 mm)
	Diameter of crankshaft connecting rod journal	ϕ 44.9 $\pm$ 0.01	
	Thickness of connecting rod bearing shell	1.48 $\begin{smallmatrix} +0.007 \\ +0 \end{smallmatrix}$	
13	Thickness of main bearing shell	2.131 $\begin{smallmatrix} +0.007 \\ +0 \end{smallmatrix}$	0.000–0.002 in (0.011–0.058 mm)
	Diameter of main bearing hole on cylinder block	ϕ 53.2935 $\pm$ 0.0065	
	Diameter of crankshaft main journal	ϕ 48.99 $\pm$ 0.01	
14	Width of crankshaft thrust surface on cylinder block	24 $\pm$ 0.03	0.004–0.012 in (0.093–0.303 mm)
	Width of crankshaft thrust surface	28.825–28.875	
	Thickness of thrust plate	2.326 $\pm$ 0.025	
15	Width of crankshaft key groove	4.75 $\pm$ 0.013	-0.002–0 in (-0.051–0 mm)
	Width of woodruff key	4.763~ 4.788	
16	Diameter of crankshaft front oil seal	ϕ 50.30– ϕ 50.50	-0.02– -0.01 in (-0.50– -0.26 mm)
	Oil seal hole on oil pump	ϕ 50 $\begin{smallmatrix} +0.04 \\ +0 \end{smallmatrix}$	
17	Diameter of crankshaft rear oil seal	ϕ 108 $\begin{smallmatrix} +0.05 \\ -0.3 \end{smallmatrix}$	-0.02– -0.01 in (-0.50– -0.25 mm)
	Outer diameter of oil seal hole on rear oil seal retainer	ϕ 108 $\begin{smallmatrix} +0.05 \\ -0 \end{smallmatrix}$	
18	Outer diameter of crankcase bleed pipe	ϕ 18 $\begin{smallmatrix} +0.3 \\ -0.2 \end{smallmatrix}$	0.01 in (0.15–0.30 mm)
	Diameter of bleed pipe hole on cylinder block	ϕ 18 $\begin{smallmatrix} +0.05 \\ -0 \end{smallmatrix}$	
19	Outer diameter of oil level indicator tube	ϕ 12 $\begin{smallmatrix} +0.05 \\ -0 \end{smallmatrix}$	0.00 in (-0.09–0.01 mm)
	Diameter of oil level indicator tube hole on cylinder block	ϕ 11.96 $\begin{smallmatrix} +0.05 \\ -0 \end{smallmatrix}$	
20	Outer diameter of valve guide	ϕ 11.04– ϕ 11.051	-0.002–0.000 in (-0.051– -0.022 mm)
	Diameter of valve guide bottom hole on cylinder head	ϕ 11 $\begin{smallmatrix} +0.018 \\ -0 \end{smallmatrix}$	
21	Diameter of exhaust valve seat ring	ϕ 25.51 $\begin{smallmatrix} +0 \\ -0.016 \end{smallmatrix}$	0.000– -0.003 in (-0.11– -0.078 mm)
	Diameter of exhaust valve seat ring bottom hole on cylinder head	ϕ 25.4 $\begin{smallmatrix} +0.016 \\ -0 \end{smallmatrix}$	

Dimensional Tolerances and Fits (page 3 of 5)

	Name	Dimension and Tolerance	Fit Clearance
22	Outer diameter of intake valve seat ring	$\phi 31.31 \begin{smallmatrix} +0 \\ -0.016 \end{smallmatrix}$	0.000–.003 in (-0.11– -0.078 mm)
	Diameter of intake valve seat ring bottom hole on cylinder head	$\phi 31.2 \begin{smallmatrix} +0 \\ -0.016 \end{smallmatrix}$	
23	Outer diameter of intake valve stem	$\phi 5.98 \pm 0.008$	
	Diameter of intake valve guide hole on cylinder head	$\phi 6 \begin{smallmatrix} +0.015 \\ -0 \end{smallmatrix}$	
24	Diameter of exhaust valve stem	$\phi 5.96 \pm 0.008$	0.001–0.002 in (0.032–0.063 mm)
	Diameter of exhaust valve guide hole on cylinder head	$\phi 6 \begin{smallmatrix} +0.015 \\ -0 \end{smallmatrix}$	
25	Width of camshaft key groove	$4.949 \begin{smallmatrix} +0.025 \\ -0 \end{smallmatrix}$	-0.002–0.000 in (-0.051–0.004 mm)
	Width of woodruff key	4.97–5	
26	Diameter of camshaft No.1 journal	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	0.000–0.002 in (0.02–0.054 mm)
	Diameter of cylinder head No.1 bearing hole	$\phi 26 \begin{smallmatrix} +0.021 \\ -0 \end{smallmatrix}$	
27	Diameter of camshaft No.2 journal	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	
	Diameter of cylinder head No.2 bearing hole	$\phi 26 \begin{smallmatrix} +0.021 \\ -0 \end{smallmatrix}$	
28	Diameter of camshaft No.3 journal	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	
	Diameter of cylinder head No.3 bearing hole	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	
29	Diameter of camshaft No.4 journal	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	
	Diameter of cylinder head No.4 bearing hole	$\phi 26 \begin{smallmatrix} +0.021 \\ -0 \end{smallmatrix}$	
30	Diameter of camshaft No.5 journal	$\phi 26 \begin{smallmatrix} +0.02 \\ -0.033 \end{smallmatrix}$	
	Diameter of cylinder head No.5 bearing hole	$\phi 26 \begin{smallmatrix} +0.021 \\ 0 \end{smallmatrix}$	
31	Outer diameter of camshaft oil seal	$\phi 52 \begin{smallmatrix} +0.021 \\ -0 \end{smallmatrix}$	0.00–0.01 in (-0.04– -0.17 mm)
	Diameter of oil seal hole on cylinder head	$\phi 52 \begin{smallmatrix} +0.03 \\ -0 \end{smallmatrix}$	

Dimensional Tolerances and Fits (page 4 of 5)

	Name	Dimension and Tolerance	Fit Clearance
32	Bore of rocker shaft on intake rocker assembly (LH)	$\phi 19 \begin{smallmatrix} +0.035 \\ +0.017 \end{smallmatrix}$	0.001–0.000 in (0.019–0.050 mm)
	Diameter of intake rocker shaft	$\phi 19 \begin{smallmatrix} -0.002 \\ -0.015 \end{smallmatrix}$	
	Bore of rocker shaft on intake rocker assembly (RH)	$\phi 19 \begin{smallmatrix} +0.035 \\ +0.017 \end{smallmatrix}$	
	Diameter of intake rocker shaft	$\phi 19 \begin{smallmatrix} -0.002 \\ -0.015 \end{smallmatrix}$	
	Exhaust rocker assembly	$\phi 19 \begin{smallmatrix} +0.035 \\ +0.017 \end{smallmatrix}$	
	Diameter of exhaust rocker shaft	$\phi 19 \begin{smallmatrix} -0.002 \\ -0.015 \end{smallmatrix}$	
33	Bore of intake rocker spacer sleeve	$\phi 19 \begin{smallmatrix} -0.2 \\ -0.3 \end{smallmatrix}$	-0.0012– -0.0007 in (-0.0298– -0.0185 mm)
	Diameter of intake rocker shaft	$\phi 19 \begin{smallmatrix} -0.002 \\ -0.015 \end{smallmatrix}$	
34	Outer diameter of spark plug socket	$\phi 25 \begin{smallmatrix} +0.101 \\ +0.088 \end{smallmatrix}$	-0.004– 0.003 in (-0.101– -0.067 mm)
	Diameter of spark plug socket hole on cylinder head	$\phi 25 \begin{smallmatrix} +0.021 \\ +0 \end{smallmatrix}$	
35	Diameter of dowel pin hole on cylinder block	$\phi 15.332 \pm 0.025$	-0.008– -0.002 in (-0.203– -0.053 mm) -0.001–0.003 in (-0.016–0.084 mm)
	Diameter of spring dowel pin on cylinder head	$\phi 15.4\text{--}\phi 15.5$	
	Diameter of dowel pin hole on cylinder head	$\phi 15.464 \pm 0.02$	
36	Outer diameter of flywheel	$\phi 250.9 \begin{smallmatrix} +0 \\ -0.02 \end{smallmatrix}$	-0.01– -0.026 in (-0.20– -0.65 mm)
	Bore of gear ring	$\phi 250.25 \begin{smallmatrix} +0.25 \\ -0 \end{smallmatrix}$	
37	Bore of dowel pin hole on flywheel	$\phi 8 \begin{smallmatrix} -0.027 \\ -0.042 \end{smallmatrix}$	-0.002–0.000 in (-0.042– -0.005 mm)
	Outer diameter of clutch dowel pin	$\phi 8 \begin{smallmatrix} +0 \\ -0.022 \end{smallmatrix}$	
38	Diameter of spring dowel pin hole on clutch	$\phi 8 \begin{smallmatrix} +0.055 \\ +0.025 \end{smallmatrix}$	0.001–0.003 in (0.025–0.077 mm)
	Outer diameter of clutch dowel pin	$\phi 8 \begin{smallmatrix} +0 \\ -0.022 \end{smallmatrix}$	
39	Diameter of bore of crankshaft gear	$\phi 28.010\text{--}\phi 28.046$	0.000–0.003 in (0.01–0.066 mm)
	Diameter of crankshaft gear shaft	$\phi 27.99 \pm 0.01$	
40	Width of crankshaft gear key groove	4.77–4.85	-0.001– -0.003 in (-0.018–0.087 mm)
	Width of woodruff key	4.763–4.788	

Dimensional Tolerances and Fits (page 5 of 5)

	Name	Dimension and Tolerance	Fit Clearance
41	Dimension of crankshaft gear key groove	30.30–30.50	0.023–0.031 in (0.581–0.781 mm)
	Dimension of key groove on crankshaft gear journal	23.65± 0.10	
	Height of woodruff key	5.842–5.969	
42	Diameter of bore of crankshaft pulley	φ 28.015–φ 28.053	0.001–0.003 in (0.015–0.073 mm)
	Diameter of crankshaft gear shaft	φ 27.99± 0.01	

Mounting Torques for Major Bolts on Engine

Mounting Torques (page 1 of 2)

	Connection Location	Description	Bolt (thread) Specification	Qty.	Tightening Torque (tighten at one drive)
1	Main bearing cap-cylinder block	Main bearing cap bolt	M12×80	9	797–885 in-lbs (90–100 Nm)
		Main bearing cap stud	M12×80 M8×18	1	
2	Plug for oil passage on cylinder block	Inner hexagon tapered plug	–	2	168–221 in-lbs (19–25 Nm)
		Inner hexagon tapered plug	–	1	204–239 in-lbs (23–27 Nm)
3	Connecting rod cap-connecting rod body	Connecting rod bolt	M8×1×40.5 40.5	8	266–319 in-lbs (30–36 Nm)
4	Water pump-cylinder block	Hexagon flange bolt	M6×1×25	4	71–97 in-lbs (8–11 Nm)
5	Oil pump-cylinder block	Hexagon flange bolt	M6×1×30	6	71–97 in-lbs (8–11 Nm)
6	Rear oil seal retainer-cylinder block	Hexagon flange bolt	M6×1×25	4	71–97 in-lbs (8–11 Nm)
7	Oil strainer-cylinder block	Hexagon flange bolt	M8×16	1	150–204 in-lbs (17–23 Nm)
8	Oil strainer-main bearing cap	Hexagon lock nut	M8	1	150–204 in-lbs (17–23 Nm)
9	Oil strainer-oil pump	Inner hexagon socket cap head screws with spring washer	M6×20	2	71–106 in-lbs (8–12 Nm)
10	Oil Pan	Hexagon bolt unit	M6×30	14	53–71 in-lbs (7±1 Nm)
		Socket head cap screw unit	M6×30	4	
11	Drain plug on oil pan	Plug	M16×1.5	1	195–248 in-lbs (25±3 Nm)
12	Intake manifold-cylinder head	Equal length stud	M8×1.25	8	71–106 in-lbs (10±2 Nm)
		Plain washer	–	6	–
		Hexagon lock nut	M8	8	142–177 in-lbs (16–20 Nm)
13	Canister solenoid valve bracket-intake manifold	Hexagon flange bolt	M8×1.25×16	1	142–177 in-lbs (18±2 Nm)
14	Intake manifold upper/lower bracket-manifold	Hexagon flange bolt	M8	0.5	142–177 in-lbs (16–20 Nm)
15	Intake manifold upper/lower bracket-cylinder block	Hexagon flange bolt	M10	1	221–257 in-lbs (25–29 Nm)
16	Exhaust manifold-cylinder head	Stud	M8×1.25	1	71–106 in-lbs (10±2 Nm)
		Equal-length stud	M8×1.25	7	71–106 in-lbs (10±2 Nm)
		Hexagon flange nut	M8	8	177–257 in-lbs (20–25 Nm)

Mounting Torques (page 2 of 2)

	Connection Location	Description	Bolt (thread) Specification	Qty.	Tightening Torque (tighten at one drive)
17	Rocker shaft-camshaft bearing cap-cylinder head	Rocker shaft bolt	M8×65	10	1st step: 119–146 in-lbs (15 ±1.5 Nm) 2nd step: 252–279 in-lbs (30±1.5 Nm)
18	Camshaft No.1 bearing cap-cylinder head	Hexagon flange bolt	M6×1× 25	2	75–93 in-lbs (9.5±1 Nm)
19	Cylinder head-cylinder block	Cylinder head bolt	M10×94	10	1st step: 177–257 in-lbs (20–25 Nm) 2nd step: 354–398 in-lbs (40–45 Nm) 3rd step: turn 90° 4th step: turn 90°
20	Oil filter adapter- cylinder block-oil filter	Adapter-oil filter	M	1	124–186 in-lbs (14–21 Nm)
21	Camshaft timing gear-camshaft	Hexagon bolt	M12×1.5×30	1	797–885 in-lbs (95± 5 Nm)
22	Tensioner-cylinder block	Hexagon flange bolt	M8×25	2	142–177 in-lbs (16–20 Nm)
23	Flywheel bolt	Hexagon flange bolt	M10×1×25	6	929–1018 in-lbs (105–115 Nm)
24	Tensioner-cylinder block	Hexagon flange bolt	M8×25	2	142–177 in-lbs (16–20 Nm)
25	Valve cover- cylinder head	Bolt	M6×25	12	53–71 in-lbs (6–8 Nm)
		Bolt	M6×40	1	
26	Camshaft position sensor-valve cover	Hexagon flange bolt	M6	1	66–75 in-lbs (8±0.5 Nm)
27	Thermostat seat and cover	Hexagon bolt M6x20	M6	4	71–97 in-lbs (8–11 Nm)
28	Thermostat seat with thermostat assembly-cylinder head	Hexagon flange bolt M6x60	M6	3	71–97 in-lbs (8–11 Nm)
29	Upper timing cover with gasket assembly-cylinder block	Hexagon flange bolt	M6×1×55	2	71–97 in-lbs (8–11 Nm)
30	Lower timing cover with gasket assembly-cylinder block	Hexagon flange bolt	M6×1×55	2	71–97 in-lbs (8–11 Nm)
31	Torsional vibration damper-crankshaft	Hexagon bolt	M12×1.5	1	929–1018 in-lbs (105–115 Nm)
32	Ignition coil-thermostat seat	Hexagon flange bolt	M6×45	4	71–97 in-lbs (8–11 Nm)
33	Coolant piping assembly-cylinder block	Hexagon bolt	M10	1	319–389 in-lbs (40±4 Nm)
34	Coolant piping assembly-cylinder head	Hexagon flange bolt	M6×16	1	62–80 in-lbs (8±1 Nm)
35	Spark plug cylinder head	Spark plug assembly	M14×1.25	–	239–292 (30±3 Nm)

Parameters to be Measured during Assembly

1	Protrusion height of crankshaft upper woodruff key	0.055–0.068 in (1.392–1.739 mm)
2	Axial clearance of crankshaft	0.004–0.012 in (0.093–0.303 mm)
3	Rotating torque of crankshaft (with piston and connecting rod assembly and thrust plate installed)	142 in-lbs (16 Nm) max (6 turns at most)
4	Rotating torque of crankshaft (with cylinder head installed but without spark plugs installed)	274 in-lbs (31 Nm) max (6 turns at most)
5	Press-in depth of bowl plug on cylinder block/cylinder head should below the plane	0.0 in (0–0.5 mm)
6	Flatness between oil pump bottom plane and oil pan flange on cylinder block	±0.01 in (± 0.26 mm)
7	Flatness between rear oil seal retainer bottom plane and oil pan flange on cylinder block	±0.01 in (± 0.26 mm)
8	During a pressure test for short cylinder block assembly, final pressure	6.2 psi (42.7 kPa)
9	During a leakage test of valve, test pressure	7.3 psi (50 kPa)
10	After inflation, leakage	100ml/min (3.4 oz/min)
11	Protrusion height of camshaft woodruff key	0.06–0.08 in (1.64–2.11 mm)
12	Axial clearance of camshaft	0.004–0.006 in (0.095–0.153 mm)
13	Clearance standard between camshaft position sensor and signal wheel of speed sensor	0.0 in (0.8–1.0 mm)
14	During an air pressure test for engine assembly, minimum pressure	5.3 psi (36.5 kPa)

Oil Sites during Engine Assembly

	Oil Site	Remark
1	Piston pin	
2	Piston ring	
3	Piston	
4	Cylinder tube hole	
5	Main bearing shell and crankshaft main journal	
6	Crankshaft thrust plate (oil groove side)	
7	Connecting rod bolt head and threaded portion	
8	Connecting rod bearing shell and crankshaft connecting rod journal	
9	Main bearing bolt head	
10	Front oil seal edge and crankshaft oil seal journal	
11	Rear oil seal edge and crankshaft oil seal journal	
12	Valve guide hole	
13	Valve stem	
14	Camshaft cam and journal	
15	Hydraulic tappet	
16	Hydraulic tappet hole and valve rocker head	
17	Valve small end and valve rocker head	
18	Valve seat ring hole (cylinder head assembly)	
19	Valve guide or guide bottom hole (cylinder head)	
20	Camshaft oil seal edge or journal	
21	Oil filter gasket surface	

NOTE: Use SL grade and above Mobile Esso 5W-40 (for winter use) and 10W-40 (for summer use) as engine oil.

Rubber Coating Positions during Engine Assembly

	Sealant Coating Position	Sealant	Remark
1	Bowl plug on cylinder block	Loctite™ 648	
2	1/4, 3/8NPTF plug	Loctite™ 243	
3	Between oil pump bottom surface and cylinder block bottom plane	Loctite™ 5699	
4	Between rear oil seal bottom surface and cylinder block bottom plane	Loctite™ 5699	
5	Mating face between oil pan gasket and oil pan semicircle	Loctite™ 5699	
6	Bowl plug on cylinder head	Loctite™ 648	
7	Camshaft No.1 bearing cap	Loctite™ 574	
8	Spark plug socket	Loctite™ 648	
9	Coolant temperature sensor	Loctite™ 243	
10	Oil pressure switch	Loctite™ 243	
11	Studs on intake and exhaust manifolds	Sealant-coat beforehand or apply Loctite™ 262 or 271 during assembly	
12	Flywheel bolt	Sealant-coat beforehand	
13	Valve cover gasket and bolt	Loctite™ 5910	

Part No. 0J1556SMNL Rev. A 09/15/15 Printed in USA
© Generac Power Systems, Inc. All rights reserved
Specifications are subject to change without notice.
No reproduction allowed in any form without prior written
consent from Generac Power Systems, Inc.



Generac Power Systems, Inc.
S45 W29290 Hwy. 59
Waukesha, WI 53189
1-888-GENERAC (1-888-436-3722)
generac.com